

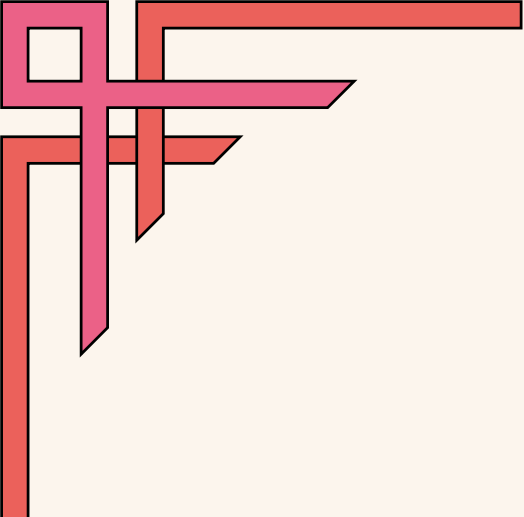
Week-4 Deliverable

**STRATEGIC RECOMMENDATIONS FOR
IMPROVING CUSTOMER SUPPORT
TICKET OPERATIONS REPORT**

Team-2

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CONTENTS

- 1. Introduction**
- 2. Dataset & Analysis Context**
- 3. Power BI Dashboard & Visuals Summary**
- 4. Executive Summary**
- 6.Strategic Recommendations for Workflow Improvements**
- 6. Freshdesk Scenario - Automation Design**
- 7. Learning Outcomes**
- 8. Conclusion**

- **Additional Attachment: "Presentation Slides (Canva View Link)"**
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1. Introduction

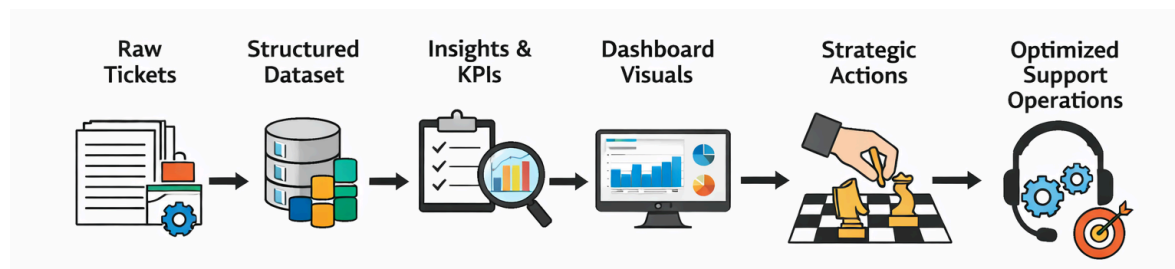
The Week 4 “**Strategic Recommendations for Improving Customer Support Ticket Operations Report**” activity focuses on consolidating analytical findings from previous weeks and translating them into actionable, data-driven strategies for enhancing customer support performance. Building on the ticket mapping, data cleaning, tagging, sentiment analysis, and dashboard-based insights developed during Weeks 1–3, this final phase emphasizes strategic thinking, executive reporting, and operational optimization.

During this week, observed trends in ticket volume, recurring issues, agent workload, response efficiency, and customer sentiment are interpreted to understand their impact on overall service quality and customer experience. The objective is not only to identify performance gaps and operational bottlenecks, but also to propose practical workflow improvements that can be implemented within a real-world helpdesk environment such as Freshdesk.

By combining analytical insights with structured recommendations, this report demonstrates how customer support data can guide smarter decision-making, improve resolution efficiency, and enhance customer satisfaction. The outcome is a concise executive-level report supported by visuals, dashboards, and workflow scenarios that illustrate how analytics-driven strategies can strengthen customer support operations.

Our aim is –

- To consolidate and interpret key trends and insights identified across Weeks 1–3.
- To connect ticket patterns, agent performance, and sentiment insights with customer experience outcomes.
- To identify workflow inefficiencies, recurring issues, and high-impact problem areas.
- To propose practical, data-backed strategies to improve ticket handling, prioritization, and resolution efficiency.
- To present findings through a clear, professional executive report supported by visuals and Freshdesk workflow scenarios.



2. Dataset & Analysis Context

2.1 Freshdesk Customer Support Tickets Dataset Overview

The analysis throughout Weeks 1–4 is based on the Freshdesk Customer Support Tickets dataset, which represents anonymized customer support interactions handled within a helpdesk environment. Each record corresponds to a unique customer issue and captures operational, time-based, and experience-related attributes such as ticket category, priority, status, tags, resolution timelines, agent assignment, and customer sentiment. This dataset serves as the foundation for ticket mapping, data cleaning, dashboard analysis, and strategic recommendations.

2.2 Dataset Reference & Continuity Across Weeks

Week	Dataset Stage (links)	Description
—	Assigned Dataset  Freshdesk Customer Support Tickets Dataset	Raw Freshdesk Customer Support Tickets dataset
1	Filled Dataset  Freshdesk Customer Support Tickets Dataset Filled	Ticket mapping with populated Category, Priority, Status, Tags, and Sentiment
2	Cleaned & Structured Dataset Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled.csv	Standardized, cleaned dataset prepared using Python
3	Visual Analytics	Cleaned & Structured Dataset used for dashboards, KPIs, and forecasting
4	Strategic Interpretation	Consolidation of insights for recommendations

This report builds directly on the outputs developed during Weeks 1–3, ensuring analytical continuity and consistency.

2.3 Ticket Mapping Methodology

To enable structured analysis, initially empty or unstructured fields were manually populated using consistent rules:

- **Category:** Assigned based on the primary issue described in the ticket (e.g., Technical, Billing, Account Management, Feature Request) to support issue-type analysis.
- **Priority:** Defined using urgency and business impact indicators to guide SLA expectations and ticket sequencing.
- **Status:** Standardized across lifecycle stages (Open, Pending, Resolved, Closed) to track workflow progress.
- **Tags:** Applied as short, descriptive labels (e.g., login-issue, server-error, payment-failed) to identify recurring problems and improve pattern recognition.
- **Sentiment:** Classified as Positive, Neutral, or Negative based on customer tone expressed in the subject and description to capture experience impact.

2.4 Data Cleaning Process

Following ticket mapping, Python-based data cleaning was performed to ensure reliability and analytical accuracy:

- Core identifiers, categories, priorities, and statuses were already consistent and required minimal correction.
- Date fields (Created Date and Resolved Date) were converted from text to datetime format to support time-based calculations.
- Minor inconsistencies were detected in the Sentiment column (e.g., extra spaces in “Neutral”) and corrected.
- Duplicate rows and missing values were checked, with none detected across the dataset.

This process resulted in a fully structured and analysis-ready dataset.

2.5 Dashboard Creation Methodology

Dashboard was developed using the cleaned dataset to translate ticket data into actionable insights:

- **Data Modeling:** Structured columns were used to support aggregations by date, category, priority, agent, and sentiment.
- **DAX Calculations:** Measures were created to calculate KPIs such as total tickets, resolution rate, average resolution time, SLA compliance, and sentiment-based CSAT proxy.
- **KPI Selection:** Metrics were chosen based on operational relevance, focusing on efficiency, workload distribution, service quality, and customer experience.
- **Visualizations & Filters:** Bar charts, line charts, donut charts, and slicers were applied to enable dynamic exploration across time periods, issue types, priorities, agents, and sentiment categories.

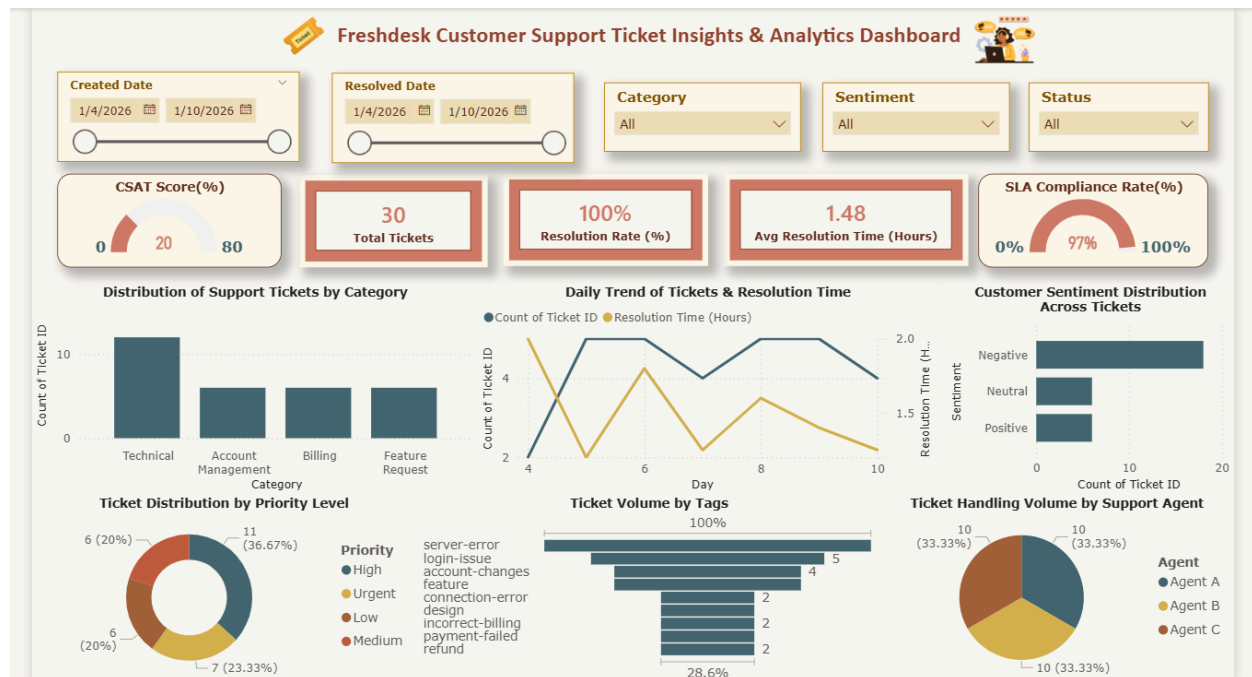
2.6 Scope and Limitations

The analysis is subject to the following scope and constraints:

- **Dataset Size:** Limited to 30 rows and 11 columns, which restricts large-scale generalization.
- **Time Range:** Covers a short period (January 4–10, 2026), limiting long-term trend and seasonality analysis.
- **Data Constraints:** No direct CSAT score is available; customer sentiment is used as a proxy for satisfaction.
- **Interpretation Scope:** Insights are indicative and best suited for demonstrating analytical and strategic methodologies rather than enterprise-wide conclusions.

3. Power BI Dashboard & Visuals Summary

3.1 Dashboard summary



The above **Freshdesk Customer Support Ticket Insights & Analytics – Power BI Dashboard** provides a comprehensive view of customer support performance for the first week of January 2026. It visually summarizes key KPIs and trends, enabling quick identification of operational bottlenecks, recurring issues, and areas impacting customer satisfaction, which will be detailed in the sections below.

Table 1: KPI Observations

KPIs	Observation
CSAT Score (20%)	The CSAT score is critically low at 20%, signaling major customer dissatisfaction despite quick ticket resolutions. It highlights a disconnect between resolution speed and service quality. This suggests agents may be closing tickets too quickly without fully resolving customer issues.
Total Tickets (30)	Shows the total volume of support requests received in the first week of January 2026. Helps understand the workload handled by the team. No major backlog is apparent, but the high-pressure environment is evident given ticket composition.
Resolution Rate (100%)	Indicates that all tickets have been closed, suggesting high operational efficiency. However, the perfect rate may mask rushed responses, compromising customer satisfaction. It raises a flag about the balance between speed and quality.
Avg Resolution Time (1.48 Hours)	Tickets are resolved extremely fast, averaging less than 90 minutes. While efficient, this may contribute to low CSAT as issues may not be thoroughly addressed. This metric indicates the team is prioritizing speed over quality.
SLA Compliance (97%)	Almost all tickets meet internal SLA deadlines, reflecting strong adherence to internal standards. Despite this, SLA is not 100% because a few complex or technical tickets slightly exceeded the target resolution time. Negative sentiment persists, showing SLA compliance alone does not guarantee customer satisfaction. Highlights a potential operational vs. experiential gap.

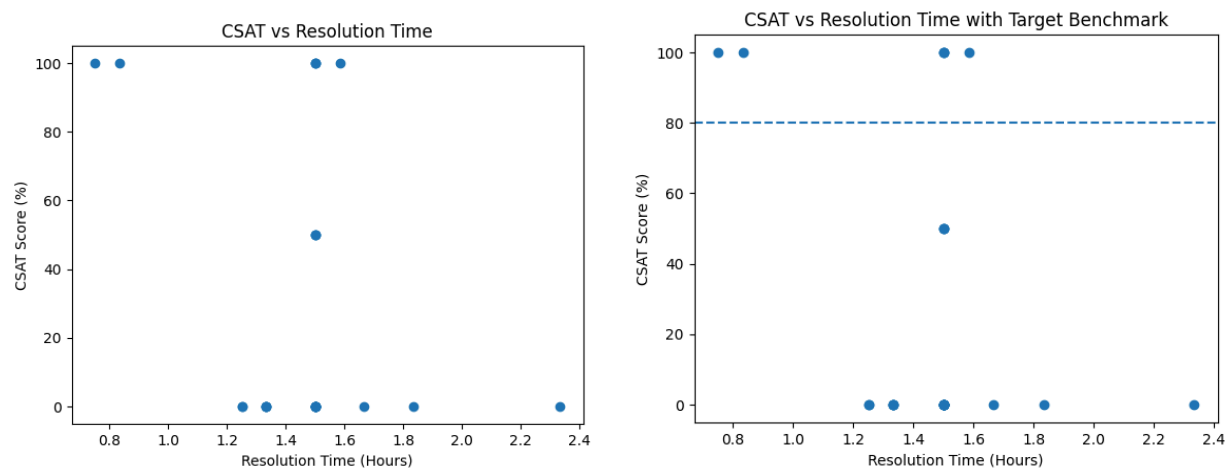
Table 2: Visualization Observations

Visualizations	Observation
Distribution of Support Tickets by Category	Technical tickets dominate, nearly double other categories like Billing or Account Management. This indicates recurring system or access-related issues are driving ticket volume. Resolving these root causes could significantly reduce overall workload and negative sentiment.
Daily Trend of Tickets & Resolution Time	Shows fluctuations in daily ticket volume and resolution speed, with spikes in resolution time on the 4th and 6th despite lower ticket counts. Suggests

	operational bottlenecks on specific days. Implies that certain technical issues may require more attention or specialized handling.
Customer Sentiment Distribution	Negative sentiment overwhelmingly dominates, with very few neutral or positive interactions. Indicates customers are unhappy even though tickets are resolved quickly. Reflects the “speed vs. quality” paradox where rapid responses fail to satisfy users.
Ticket Distribution by Priority	Over 60% of tickets are High or Urgent, creating a high-pressure environment for agents. Suggests that operational stress is significant, which may impact service quality. Continuous high-priority workload could lead to agent fatigue and reduced support effectiveness.
Ticket Volume by Tags	Funnel chart shows “server-error” and “login-issue” as the most frequent technical problems. Indicates recurring, systemic technical issues that the support team cannot fully resolve. Highlights the need for proactive product-side fixes to reduce repeated complaints.
Ticket Handling Volume by Agent	Tickets are perfectly evenly distributed among Agents A, B, and C (33.33% each). While balanced, it ignores ticket complexity and agent expertise, potentially reducing the quality of responses. Suggests mechanical assignment may contribute to low CSAT despite high resolution efficiency.

3.2 Additional Visuals Summary

CSAT vs Resolution Time - SCATTER PLOT



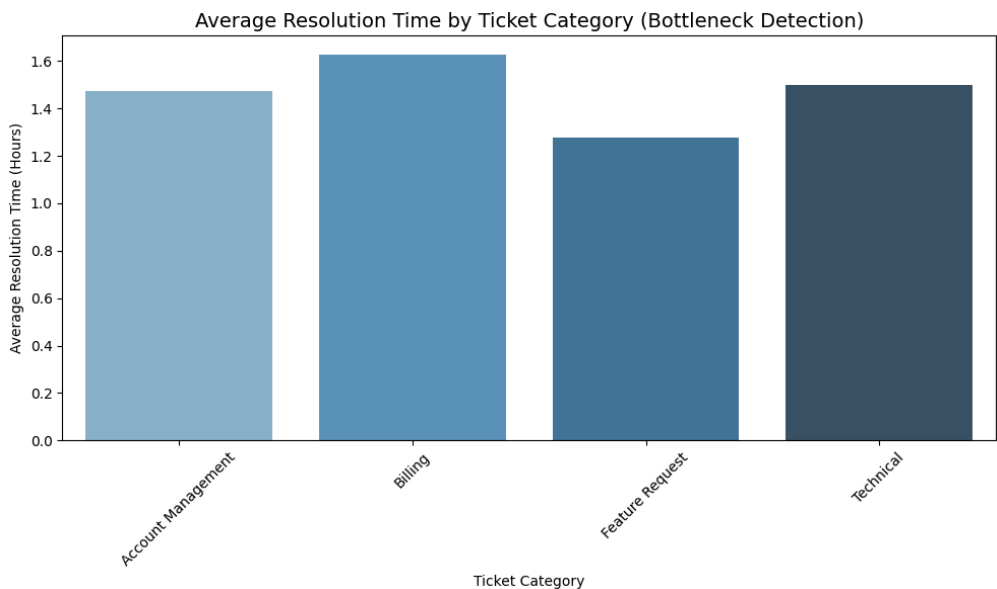
The scatter plot visualizes the relationship between ticket resolution speed and customer satisfaction (CSAT). Each dot represents an individual ticket, with the X-axis showing the **Resolution Time (Hours)** and the Y-axis showing the **CSAT Score (%)**, mapped from sentiment (Positive = 100%, Neutral = 50%, Negative = 0%). A horizontal dashed line marks the **target CSAT of 80%**, providing a benchmark for performance.

The plot reveals a **negative correlation**: as resolution time increases, CSAT generally decreases. Most tickets cluster in the **top-left “Success Zone”**, where issues are resolved quickly with high satisfaction. A few outliers in the top-right indicate “Patient Advocates” — customers who experienced longer resolution times but remained satisfied, likely due to solution quality. Conversely, tickets in the bottom-left suggest potential **quality issues**, where quick resolutions did not meet customer expectations.

These charts combined answer a fundamental question: **"How much does speed matter to our customers?"** If the CSAT line drops sharply as resolution time increases, your customers value speed above all else. If the CSAT stays steady even when time increases, your customers may value the *quality* or *accuracy* of the fix more than a quick reply.

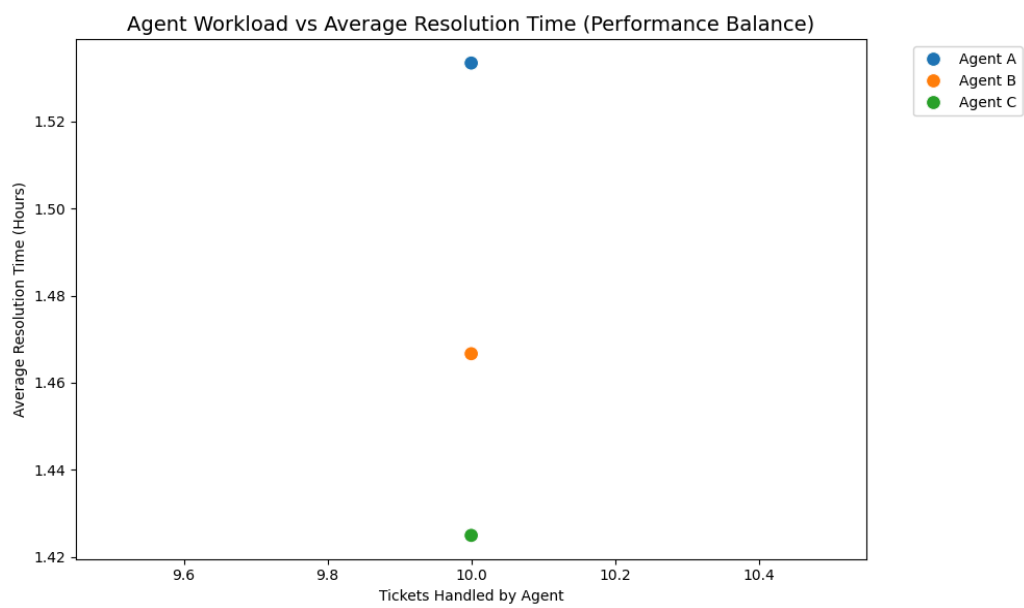
This visualization highlights the critical impact of resolution speed on customer satisfaction and identifies **bottlenecks or friction points** where delayed responses may reduce CSAT. It serves as a guide for operational improvements, such as prioritizing tickets likely to exceed the 24–48 hour threshold or enhancing resolution quality for fast-tracked tickets.

Category vs Resolution Time (Bottleneck Detection) - BAR CHART



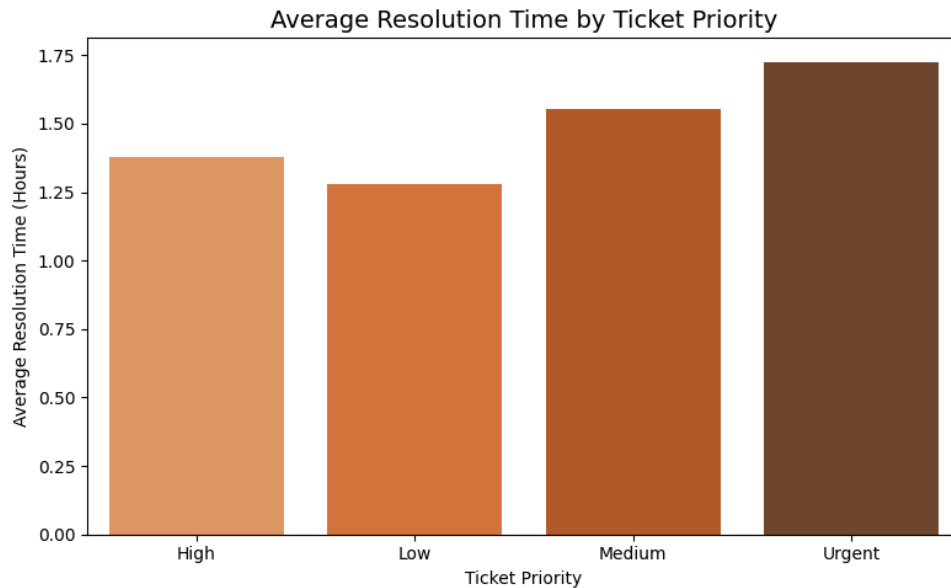
This bar chart visualizes **average resolution time by ticket category** to identify operational bottlenecks across the support process. Each bar represents a ticket category, with the **X-axis** showing the categories (Account Management, Billing, Feature Request, Technical) and the **Y-axis** showing **Average Resolution Time (Hours)**. The chart reveals that **Billing tickets have the highest resolution time (~1.6 hours)**, followed closely by **Technical tickets (~1.5 hours)**, while **Feature Requests are resolved fastest (~1.25 hours)**. Account Management tickets fall in between (~1.45 hours). These differences highlight **performance disparities and potential bottlenecks**. The slow resolution in Billing may result from complex payment processes, manual approvals, or insufficiently specialized agents. Technical tickets indicate complexity or capacity constraints, suggesting the need for better resource allocation or process improvements. The nearly **23% gap between the fastest and slowest categories** emphasizes the importance of balancing efficiency across all support areas.

Agent Workload vs Resolution Time (Performance Balance) - SCATTER PLOT



This scatter plot visualizes **average resolution time against the number of tickets handled by each agent**, highlighting individual performance balance. The **X-axis** represents the **Tickets Handled**, while the **Y-axis** shows **Average Resolution Time (Hours)**. Each point corresponds to a specific agent. The chart shows that although all three agents (A, B, and C) handled the **same number of tickets (10 each)**, there are **noticeable differences in efficiency**. **Agent C (Green)** is the fastest at ~1.425 hours, **Agent B (Orange)** averages ~1.465 hours, and **Agent A (Blue)** is the slowest at over 1.53 hours. This visualization uncovers key issues in performance management: a **0.1-hour gap between fastest and slowest agents** can accumulate into lost productivity, **Agent A underperforms**, and the **rigid round-robin workload assignment** does not leverage agent efficiency effectively. The chart suggests a need for **dynamic workload allocation** or **targeted training** to balance speed and productivity across the team.

Priority vs Resolution Time - BAR CHART



This bar chart displays **average resolution time across different ticket priority levels**, highlighting how effectively the support team addresses urgent versus lower-priority requests. The **X-axis** represents **Ticket Priority** (Low, Medium, High, Urgent), and the **Y-axis** shows **Average Resolution Time in Hours**. The chart reveals that **Urgent tickets take the longest to resolve (~1.725 hours)**, followed by Medium (~1.55 hours), High (~1.375 hours), and Low (~1.3 hours). Ideally, Urgent and High priority tickets should be resolved fastest, but the data indicates a **priority inversion**, where less critical tickets are closed quicker than the most pressing issues. Key issues identified include **critical delays in resolving urgent tickets**, potential **process inefficiencies**, and a **misalignment in workload focus**, suggesting the team may be prioritizing easier tasks over high-stakes issues. The visualization emphasizes the need for **improved triage, workload management, and escalation protocols** to ensure that critical tickets receive timely attention.

Overall, the dashboard & additional visuals highlight critical insights into ticket trends, agent performance, and customer sentiment, clearly revealing areas of strength and key challenges that require targeted improvements to enhance support efficiency and satisfaction.

→ To see how the additional visuals were created in python, click here (link):

[Additional_Ticket_Visuals.ipynb](#)

4. Executive Summary

4.1 High-Level Overview of Key Trends Identified Across Weeks 1–3

Across Weeks 1 to 3, the analysis reveals consistent patterns in customer support operations driven by recurring technical and billing issues. Ticket volume is heavily concentrated in problem-driven categories, particularly Technical and Billing, while resolution speed and SLA compliance remain high. Despite strong operational efficiency indicators such as a 100% resolution rate and 97% SLA compliance, customer sentiment remains predominantly negative, indicating a misalignment between internal performance metrics and actual customer experience. Agent workloads are evenly distributed, but performance variations and rigid assignment mechanisms limit service quality optimization.

4.2 Major Recurring Issues and Support Challenges Identified

- **Recurring technical failures (server errors and login issues)**
Affects: Customer experience, customer satisfaction, support operations
Impact: These issues prevent users from accessing core services, resulting in repeated complaints, dominant negative sentiment, and increased ticket volume that overwhelms the support team.
- **Billing-related delays and complexity**
Affects: Customer satisfaction, SLA risk, operational efficiency
Impact: Billing tickets show the highest average resolution time, increasing frustration for customers and raising the risk of SLA breaches due to manual verification and approval processes.
- **Severely low CSAT despite high SLA compliance**
Affects: Customer experience, service quality perception
Impact: A CSAT score of 20% indicates customers feel dissatisfied even when tickets are resolved quickly, showing that speed-focused resolution is failing to meet user expectations.
- **Quality vs. speed imbalance in ticket handling**
Affects: Customer satisfaction, first-contact resolution quality
Impact: Rapid ticket closures lead to incomplete or surface-level responses, causing customers to remain dissatisfied and potentially re-open or re-submit tickets.
- **Priority inversion in ticket resolution**
Affects: Escalation effectiveness, SLA risk, operational focus
Impact: Urgent tickets take longer to resolve than lower-priority ones, delaying critical issue resolution and weakening the effectiveness of the escalation framework.

- **Rigid, mechanical ticket assignment across agents**
Affects: Agent efficiency, service quality, operational optimization
Impact: Equal ticket distribution ignores agent expertise and ticket complexity, leading to slower resolutions for complex issues and inconsistent service outcomes.
- **High proportion of High and Urgent priority tickets**
Affects: Agent workload, burnout risk, support sustainability
Impact: Sustained exposure to high-priority issues increases operational pressure, encouraging rushed resolutions and reducing long-term service quality.
- **Limited automation and self-service mechanisms**
Affects: Support operations, efficiency, agent productivity
Impact: Simple, repetitive queries consume agent time, reducing capacity for complex technical and billing issues and increasing overall workload.

4.3 Overall Impact on Customer Satisfaction and Operational Efficiency

- **Which recurring issues have the highest impact on customers?**
Recurring technical failures, particularly login issues and server-related errors, have the most severe impact on customers as they directly disrupt system access and service usability. These issues create immediate frustration, reduce trust in platform reliability, and negatively affect customer experience. Billing-related issues also show a high impact due to their financial implications and perceived sensitivity, often leading to dissatisfaction when resolution is delayed. Together, these categories significantly influence customer satisfaction and increase escalation risks within support operations.
- **Are there bottlenecks in the workflow that delay resolution?**
Clear workflow bottlenecks are observed within the Technical and Billing categories, where tickets consistently record longer resolution times. These delays are primarily caused by issue complexity, dependency on backend teams, and limited first-contact resolution. Additionally, priority misalignment contributes to operational inefficiencies, as some urgent tickets are not escalated promptly. This impacts SLA compliance, increases resolution backlog, and weakens overall support operational efficiency.
- **How do agent performance patterns affect service quality?**
While ticket volume is relatively evenly distributed across agents, significant variation exists in resolution speed and effectiveness. Some agents consistently resolve tickets faster, while others take longer due to experience gaps or issue-type mismatches. The current round-robin assignment approach does not account for agent specialization, resulting in inefficient resource utilization. This directly affects service quality, prolongs resolution time, and indirectly lowers customer satisfaction despite balanced workloads.
- **Which metrics correlate most with low CSAT scores?**
Low CSAT scores correlate strongly with negative sentiment, repeated ticket occurrences, and issues categorized under Technical and Billing tags. High-priority

tickets that remain unresolved for extended periods also contribute to dissatisfaction. In contrast, resolution time and SLA compliance show a weak correlation with CSAT, indicating that customers value effective problem resolution and communication more than speed alone. This highlights a gap between operational performance metrics and perceived customer experience.

Overall, the executive analysis identifies a critical disconnect between operational efficiency and customer experience. While internal KPIs reflect strong performance, recurring technical failures, billing complexities, misaligned priorities, and rigid workflows continue to drive customer dissatisfaction. These findings highlight the need for strategic workflow optimization, improved prioritization, smarter automation, and proactive issue resolution.

So , we have outlined targeted strategic recommendations in the next section to address these challenges and improve both customer satisfaction and operational effectiveness.

5. Strategic Recommendations for Workflow Improvements

Over the 4 weeks, we identified numerous weaknesses affecting our workflow. This section will focus more on those weaknesses and find ways to improve them.

5.1 CSAT Score Calculation Method

Issue: Our method for calculating CSAT scores is highly inaccurate, as we use a customer's sentiment towards an issue rather than their satisfaction with the customer support provided. This is our only way to calculate it, however. Sentiments are a great way to identify what problems may frustrate users the most, but it's not a reliable way to calculate CSAT score.

Recommendation: A way to tackle this issue would be to introduce a short "rate your experience" survey for customers after an issue is resolved to allow users to express their satisfaction on a 5-point scale. Adding a small section where customers can briefly mention what they did/didn't like about the service provided would open doors to even more improvement.

Expected outcome: CSAT score is an extremely important measure of effectiveness. Once we have a reliable method to measure it, we can track progress and make meaningful improvements to our customer service system.

5.2 Recurring Technical Failures

Issue: Technical issues are reported very frequently and often have a negative sentiment attached to them. These issues clearly frustrate customers, but they also cause delays in operations, as a smoothly running app is vital to communication between a user and a server.

Recommendation: A suggestion to help combat this would be to frequently test the server for bugs and errors, then constantly work on fixing them. This could mean allocating more resources to the technical team as well, since expanding the team may be needed.

Expected outcome: Once the website runs smoothly, customers will feel happier using it, and their loyalty and trust would be gained.

5.3 Billing-Related Delays and Complexity

Issue: Tickets in the billing category have the highest average resolution time. This leads to a multitude of problems, including reduced CSAT score, operational issues, and a lack of trust from customers.

Solution: These issues are highly sensitive and clearly require specialized agents, their expertise will also lead to a lower resolution time. The billing team may be struggling to keep up with the increasing billing-related queries, slowing down operations. It may be helpful to expand the billing team.

Expected outcome: Increased efficiency and higher CSAT. Billing issues can particularly draw customers away, and so making the support timeline smoother will definitely increase customer loyalty. Operations will run much more smoothly with no financial delays.

5.4 Unstrategic Ticket Handling

Issue: Urgent and high priority tickets take far too long to solve in comparison to low and medium priority tickets. This shows we aren't properly prioritising what we need to. Tickets are assigned to agents randomly, ignoring the agent's skill level. These two issues show that we don't approach tickets in a well-thought-out strategic way.

Solution: Create a straightforward system to handle tickets according to their priority. Set time limits on tickets according to their priorities. Make urgent/high priority tickets appear at the top for agents to handle first, with tickets nearing their deadline appearing first within their section. Urgent/high priority tickets may be sensitive and require agents with specific skills. It would help

reduce resolution time if tickets were assigned according to an agent's expertise and experience.

Expected outcome: Critical tickets are given more importance and are handled with more urgency and care as needed. Efficiency is improved, and in turn, CSAT increases.

5.5 Potential Risks of Agent Burnout

Issue: We have a high volume of simple, repetitive tickets that could easily be automated, yet we lack self-service mechanisms, so these tickets need to be handled by agents. A large number of tickets are assigned high or urgent priority as well, adding pressure onto agents. These issues lead to an increased workload and greater pressure, risking agent burnout and sacrificing service quality.

Solution: Introducing customer automation, such as featuring FAQs, introducing a suggestions form, adding a chatbot, and implementing a self-service portal will mean customers will not need to create tickets for these simple issues. This means a significantly reduced workload for agents. Many of these simple tickets were assigned low/medium priority. With these tickets gone, the method for assigning priorities will change to adapt to the higher percentage of important tickets, and a smaller number of tickets will be assigned high/urgent priority. Agents will have more time due to the reduced number of tickets, helping tackle the speed vs quality imbalance.

Expected outcome: Reduced customer frustration as they don't have to go through the hassle of opening a ticket for simple issues and can instead solve their issues themselves. Increased efficiency. Higher CSAT due to agents working on resolving tasks with greater care and less pressure.

6. Freshdesk Scenario - Automation Design

- **Trigger Rule: New Ticket Created**

Whenever a new support ticket is submitted, Freshdesk automatically initiates an automation workflow to ensure faster handling and structured processing.

- **Auto-Detection of Recurring Issues (Tagging + Category Assignment)**

If the ticket contains common keywords such as *login failure*, *server down*, *payment error*, or *refund request*, the system assigns predefined tags and categories instantly, reducing manual effort.

- Priority Assignment for High-Impact Problems**
 Tickets tagged as **server-error** or **payment-failed** are automatically marked as **High/Urgent priority**, since these recurring issues were linked with strong negative customer sentiment earlier.
- Smart Ticket Routing to Specialized Agents**
 Freshdesk routes tickets based on category:
 Technical issues → Technical Support Agent,
 Billing/refund issues → Finance Support Agent.
 This ensures expert handling and improves resolution speed.
- Automated First Response Using Macros**
 For frequent issues like login problems or refund queries, an automatic acknowledgment is sent immediately with basic troubleshooting steps or expected timelines, improving customer reassurance.
- Escalation Workflow for SLA Risk Tickets**
 If a high-priority ticket remains unresolved beyond a set threshold, Freshdesk escalates it to a senior agent automatically, preventing SLA breaches and service delays.
- Efficiency Impact and Operational Benefit**
 This workflow reduces response time by eliminating manual classification, improving routing accuracy, and ensuring quicker attention to recurring technical and billing complaints.
- Connection to Dashboard Findings**
 The automation directly addresses the most common recurring problems identified earlier: *login issues*, *server errors*, and *payment/refund failures*, thereby improving both service consistency and customer experience.

6.1 Freshdesk Scenario: Automation Design (Workflow Rule)

Following is a structured Freshdesk automation scenario designed to reduce delays and improve support efficiency, especially for recurring technical and billing problems.

CASE: New Ticket Created

Whenever a customer submits a new ticket, Freshdesk automatically scans the subject line, category, and tags (e.g., login-issue, server-error, payment-failed).

Step 1: Auto-Tagging & Category Detection

If keywords such as login failure, server down, payment issue, refund request appear, the system assigns the appropriate tag and category instantly.

This prevents manual misclassification and speeds up routing.

Step 2: Priority Assignment Rule

Tickets tagged as server-error or payment-failed are automatically marked as High/Urgent priority, since these issues have shown strong negative sentiment impact.

Lower-impact feature requests remain Medium or Low.

Step 3: Smart Ticket Routing to Specialized Agents

Freshdesk assigns tickets based on issue type:

Technical issues → Technical Support Agent

Billing/Refund issues → Finance Support Agent

This reduces workload imbalance and ensures faster expert handling.

Step 4: Automated Customer Response (Macro Reply)

For recurring issues like login problems or refund requests, an instant acknowledgment is sent with troubleshooting steps or expected timelines.

This improves customer reassurance while the agent works on resolution.

Step 5: SLA-Based Escalation Trigger

If any high-priority ticket remains unresolved beyond a fixed threshold, Freshdesk escalates it to a senior agent automatically to prevent SLA breaches.

6.2 Efficiency and Impact on Response Time

- This automation reduces response delays by eliminating manual sorting and tagging.
- Recurring issues such as login failures, server errors, and billing complaints are addressed faster through priority escalation.
- Automated replies improve customer experience immediately, even before agent intervention.
- Overall, the workflow strengthens operational efficiency and reduces the negative sentiment caused by repeated unresolved issues.

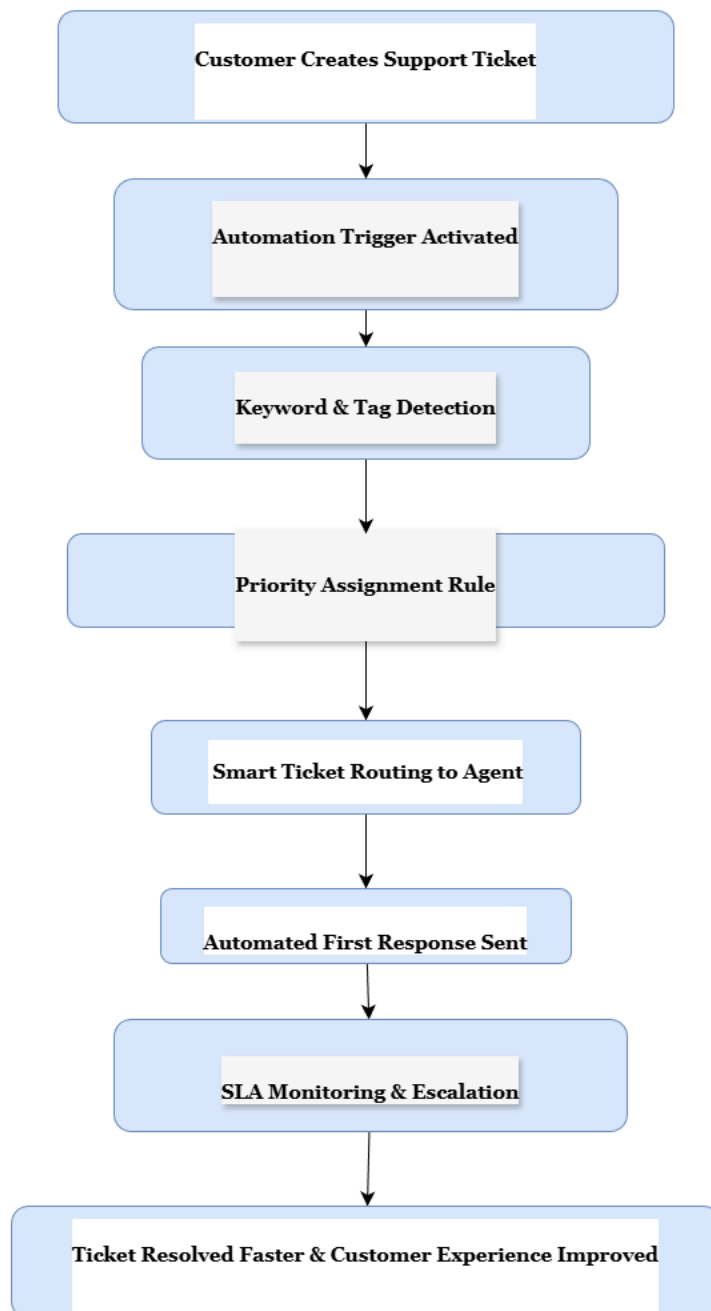
6.3 Link to Recurring Issues Identified Earlier

This scenario directly targets the most frequent problem patterns found in dashboard insights, particularly:

- **Login issues**
- **Server failures**
- **Payment and refund complaints**

By automating handling of these high-impact categories, Freshdesk support operations become faster, more consistent, and more customer-focused.

Freshdesk Ticket Automation Workflow for Recurring Issues



7. Learning Outcomes

Week 1, Data Understanding and Ticket Mapping

- Learned about the structure of customer support tickets and real-life helpdesk data from Freshdesk.
- Learned how to manually map and fill in missing information like Category, Priority, Status, Tags, and Sentiment, following a set of business rules.
- Learned to identify types of issues from unstructured descriptions in support tickets.
- Learned to pay closer attention to detail and classify operational data in a logical way.
- Learned how to effectively label support tickets to facilitate further analysis and decision-making.

Week 2, Data Cleaning and Preparation

- Learned to apply Python programming skills to clean and manipulate real-life datasets.
- Learned to identify and resolve inconsistencies in text-based data, e.g., sentiment.
- Learned to transform date-based fields into appropriate formats.
- Learned to validate data to identify potential issues like duplicate and missing values.
- Learned to recognize the value of clean and organized data in effective analytics and dashboards.

Week 3, Dashboarding and Analytical Insights

- Acquired practical skills in creating Power BI dashboards from cleaned datasets.
- Learned to create and compute KPIs such as CSAT proxy, resolution rate, SLA compliance, and average resolution time.
- Developed skills in the art of visual storytelling using charts, KPIs, slicers, and filters.
- Explored the relationships between resolution time, priority, agent performance, and customer sentiment.
- Enhanced skills in extracting useful operational insights from raw data.

Week 4, Strategic Thinking and Real-World Application

- Learned to integrate multi-week analysis into an executive-level report.

- Developed strategic thinking skills by integrating analytical insights with workflow optimization and recommendations.
- Designed a Freshdesk automation case to handle recurring issues, SLA risks, and inefficiencies.
- Acquired skills in integrating support analytics with customer satisfaction and operational efficiency.
- Enhanced skills in professional reporting, formal documentation, and data-driven decision-making.

8. Conclusion

We consolidated 4-week analysis of Freshdesk support data in this report to identify key operational gaps, including recurring technical and billing issues, priority misalignment, and a disconnect between resolution speed and customer satisfaction. We have proposed recommendations and automation workflows that focus on improving ticket prioritisation, routing accuracy, and resolution quality. Implementing these strategies is expected to result in faster and more effective resolutions, improved Customer Satisfaction Score (CSAT), and a more balanced agent workload.

Overall, we demonstrated readiness to apply analytics-driven insights to real-world customer support decision-making. Going forward, we emphasise that continuous monitoring and iterative optimisation will be essential to sustaining service quality and driving long-term customer satisfaction.

→ **Presentation Link Attachment (Canva view link):** [CID Team-2 Presentation](#)

End of Report

Thank You!